

Quiz # 01 (A)

Course Title: Peripherals, Interfacing & Embedded Systems
Course ID: CSE - 4619

Total Marks: 15

Time: 25 Mins.

Student Id:

Obtained Marks:

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- Q1.** Write short notes on fixed and variable addresses. Explain the memory-mapped I/O concept using a scenario where port # 05h is connected with a keyboard and port # 02h is connected with a 7-segment display. **3+2**
- Q2.** Suppose, it is given $V_{in} = 0.6 \text{ Volt}$, $V_{ref} = 1 \text{ Volt}$ and **8-bit** of resolution for a *Successive Approximation A/D* converter. Find the **8-bit** digital output for the given V_{in} . **3+2**
- Q3.** What is the advantage of R/2R ladder DACs over those that use binary weighted resistors? **3+2**
An 8-bit D/A converter produces $V_{out} = 0.25 \text{ V}$ for a digital input of 10000001. Find the value of V_{out} for an input of 11110000.
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